P03

Spectral Properties of the Graph Laplacian and Network Diffusion

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Published	Spectral Graph Theory (AMS) (1997)
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Read it	https://doi.org/10.1090/cbms/092 - Open access (verified excerpt)
Domain	DOM_MATH

Abstract (quoted verbatim from the publication)

The graph Laplacian L = D - A acts as a discrete Laplace-Beltrami operator. Heat diffusion across network nodes du/dt = -L*u relaxes at rates dictated by the spectral gap and algebraic connectivity.

Abstract quoted from F. R. K. Chung et al., Spectral Graph Theory (AMS) (1997).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?

This card is a workshop wrapper prepared by the organisers. Only the abstract is quoted from the publication; read the paper itself through the link above before asserting any relationship.